Task 5: Exploratory Data Analysis (EDA)

Employee Attrition Dataset — Findings Report

Objective: Extract insights using visual and statistical exploration.

Tools: Python, Pandas, Matplotlib and Seaborn.

Dataset: Synthetic practice data; it does not represent real employees.

1. Dataset overview

The dataset contains 500 rows and 7 columns: Age, Department, YearsAtCompany, MonthlyIncome, OverTime, JobSatisfaction and Attrition.

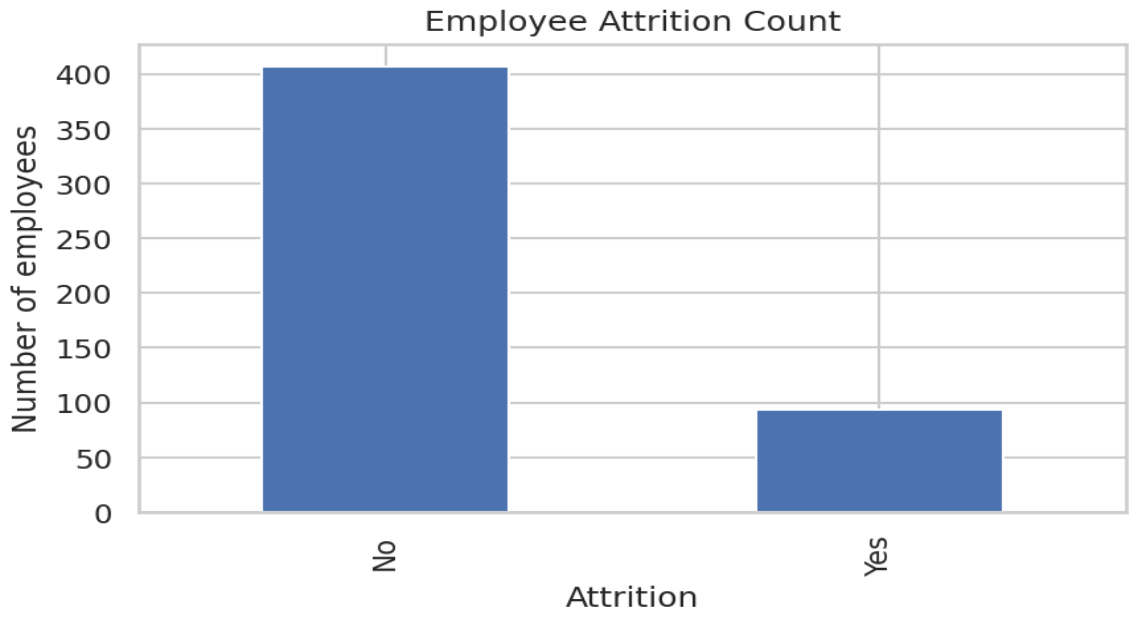
Missing values: 0. Duplicate rows: 0.

2. Statistical summary

Measure	Age	YearsAtCompany	MonthlyIncome	JobSatisfaction
count	500.0	500.0	500.0	500.0
mean	40.46	6.66	3619.56	3.03
std	11.44	4.33	1318.19	0.95
min	21.0	0.0	1800.0	1.0
25%	31.0	3.0	2618.75	2.0
50%	40.0	6.0	3458.5	3.0
75%	50.25	9.0	4375.75	4.0
max	60.0	21.0	8070.0	5.0

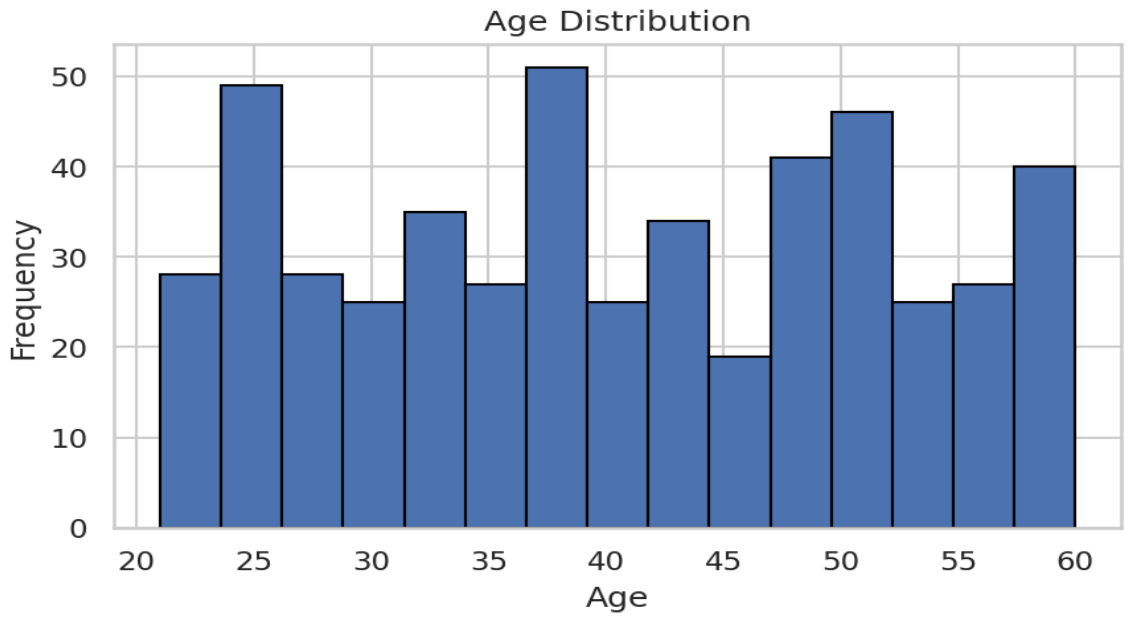
3. Visual exploration and observations

Employee attrition count



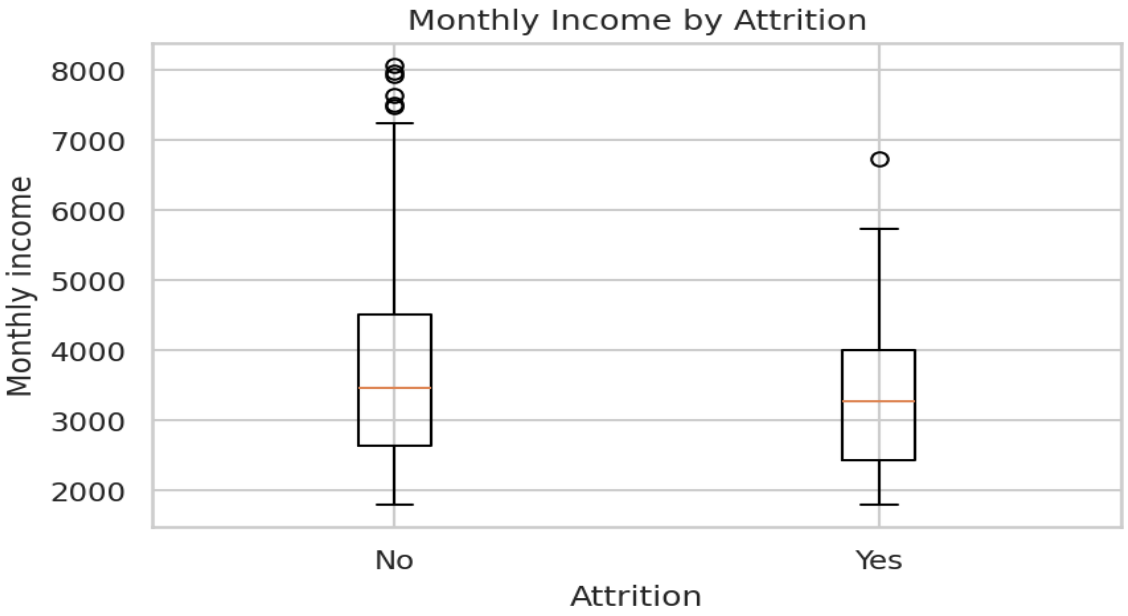
Compares employees who stayed with employees who left.

Age distribution



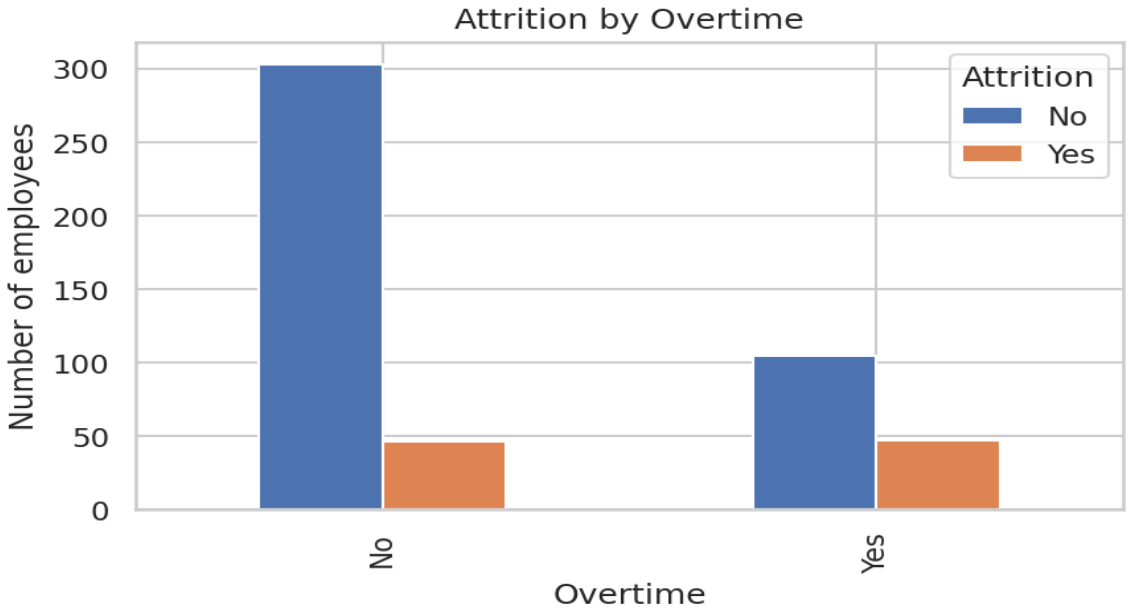
Shows how employee ages are distributed.

Monthly income by attrition



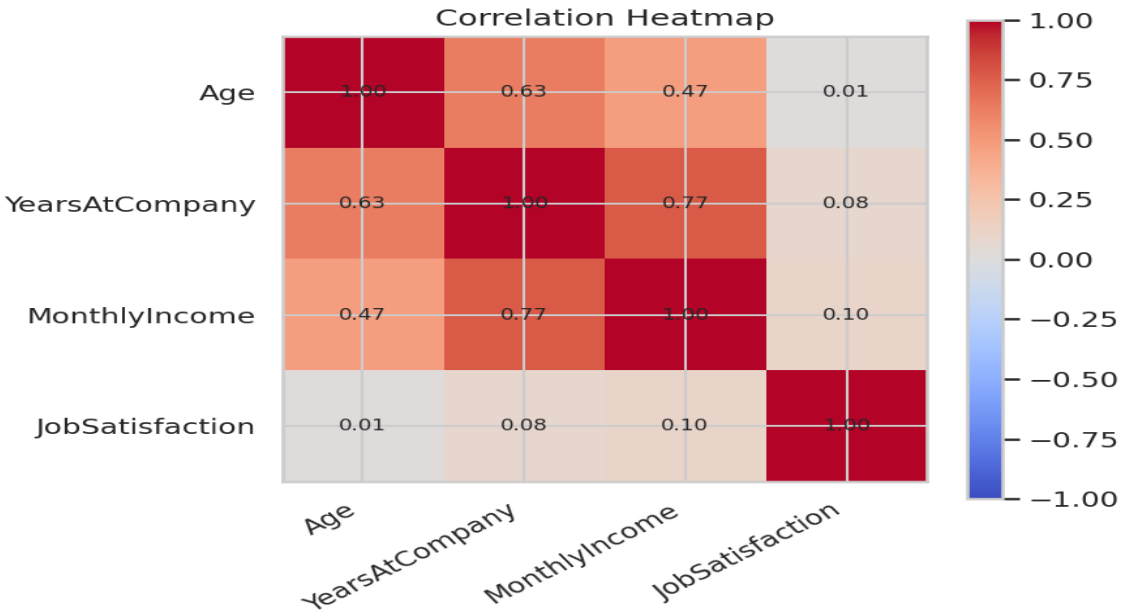
Compares income spread for employees who stayed and left.

Attrition by overtime



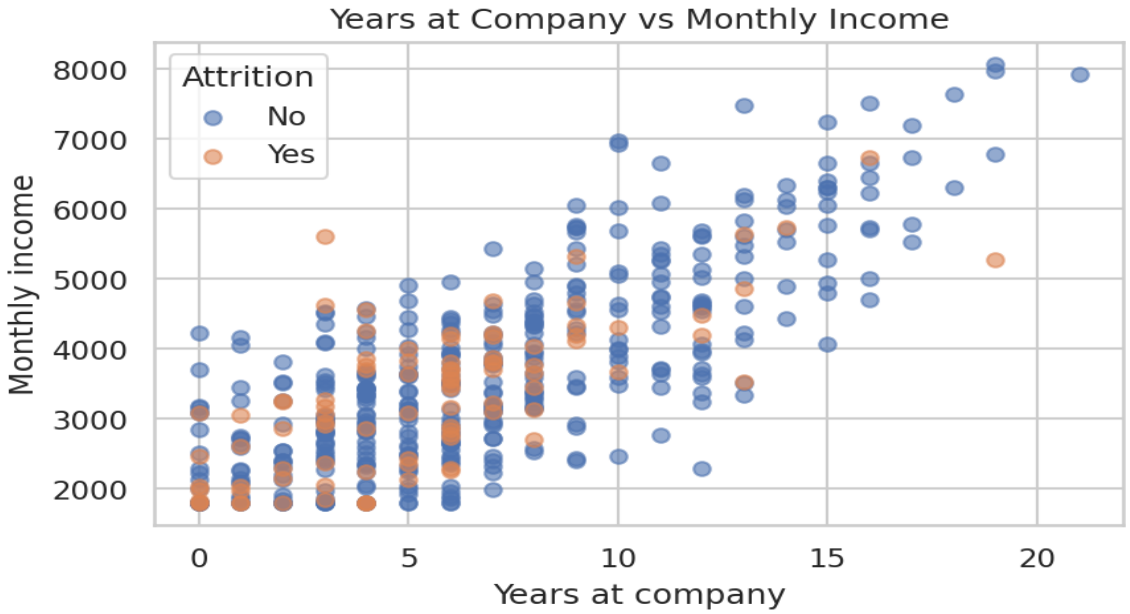
Compares attrition counts for employees with and without overtime.

Correlation heatmap



Shows correlations among numeric variables; correlation does not prove causation.

Years at company vs monthly income



Helps inspect the relationship between tenure and income.

4. Key findings

• Attrition rate in this synthetic sample: 18.6%.

• Overtime and job satisfaction are useful variables to compare with attrition.

• Income and years at the company can be explored to understand employee patterns.

• The generated dataset has no missing values or duplicate rows.

5. Conclusion

EDA uses statistics and charts to understand data before modelling. Univariate analysis studies one variable, bivariate analysis studies two variables, and multivariate analysis studies several variables together. These findings are for practice because the dataset is synthetic.